

M.A	Score
Final	✓
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N.B - M.A Mode of Assessment

Parent's Signature _____

Full Name _____ Grade & Section 9 R.No _____

Subject:-**chemistry** Year:- 2018 E.C Exam Room _____ Time allowed:- 2:00'

Portion Coverage (1-87)

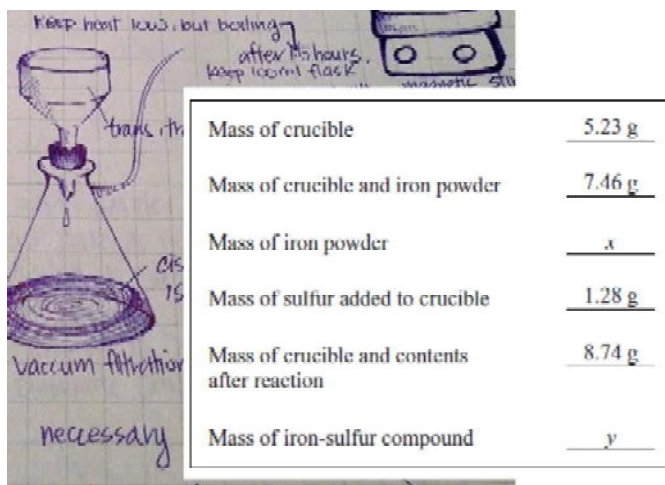
I. Write true of the statement is correct and false for the incorrect one.

1. Chemistry and Biology are related by the metabolic processes occurring inside living matter
2. The first person who discovered the electron as a sub-atomic particle and its charge to mass ratio was James Chadwick.
3. Empedocles believed that all matter was composed of only one element, which was water.
4. While adding or subtracting a quantity, the answer contains no more decimal places than the least accurate measurement.
5. Dalton's atomic theory included the concept of isotopes.

II. Chose the most correct answer from those alternatives.

- 1) According to John Dalton's atomic theory, atoms are:
 - (a) Indivisible and indestructible particles
 - (b) Made up of protons, neutrons, and electrons
 - (c) Composed of positively charged protons and negatively charged electrons
 - (d) Always in motion around the nucleus
- 2) Who discovered the electron and proposed the plum pudding model of the atom?
 - (a) Ernest Rutherford
 - (b) J.J. Thomson
 - (c) Neils Bohr
 - (d) Marie Curie
- 3) What does the term 'isotope' refer to in atomic theory?
 - (a) Atoms with different numbers of protons and electrons
 - (b) Atoms with the same number of protons but different numbers of neutrons
 - (c) Atoms with the same number of neutrons but different numbers of protons
 - (d) Atoms with different numbers of electrons in their outermost shell

- 4) A page from a lab notebook is shown below. The page displays data needed to find the empirical formula of a compound.



Calculate the values for x and y.

- x = 2.23g, y = 7.46g
 - x = 2.23g, y = 3.51g
 - x = 7.46g, y = 8.74g
 - x = 25.41g, y = 10.02g
- 5) The symbols for units of length in order from [LARGEST, ..., SMALLEST] are
- m, cm, mm, km
 - mm, m, cm, km
 - km, mm, cm, m
 - km, m, cm, mm
- 6) What happens to a decimal place in the number when you are converting from a smaller metric unit to a larger unit?
- converting doesn't require a placement of a decimal point
 - the decimal place moves to the left
 - the decimal place moves to the right
 - the decimal place does not move at all
- 7) A woman has a body mass of 53 kg. What is her mass in grams?
- 5.3
 - 0.053
 - 53
 - 53000
- 8) What is the density of a sample of mineral oil if 250mL has a mass of 230g?
- 20 g/mL
 - 0.92 g/mL
 - 1.09 g/mL
 - 57500 g/mL
- 9) Which correctly describes this student's experimental data? (The accepted value is 15.63)

	Measurement
1	12.84
2	13.02
3	12.96

- accurate but not precise
- precise but not accurate
- both accurate and precise
- neither accurate nor precise

- 10) Which one of the following results in the discovery of proton
 a) Cathode rays (b) canal rays (c) X-rays (d) alpha rays.
- 11) "Like atoms are identical in all respects". This statement of Dalton's Atomic Theory was contradicted. What discovery did contradict this?
 (a) Sub-atomic particles (b) isobars (c) isotopes (d) atomic nucleus
- 12) Canal ray experiment lead to the discovery of _____.
 a) protons b) neutrons c) electrons d) nucleus
- 13) Which of the following are properties of Anode rays? Anode rays:
 a) Travel in non-straight lines.
 b) Deflected in electric and magnetic field in opposite to that of cathode rays.
 c) Are simply negatively charged gaseous ions.
 d) Represent electrons of all atomic species.
- 14) Carbon dioxide (CO₂) is composed of one carbon atom and two oxygen atoms. Therefore, by mass, carbon dioxide can be described by the fixed ratio of :
 12 (mass of carbon) : 32 (mass of oxygen) or simplified as 3:8. this is the application of ;
 a) Law of conservation of mass c) Lussac's law
 b) Law of definite proportion d) Law of Multiple proportion
- 15) The number 5.678×10^5 is equivalent to:
 a) 0.00005678 b) 567,800 c) 56,780 d) 5,678,000
- 16) Which of the following is the SMALLEST?
 a) 12.7 cm b) 12.7 m c) 12.7 mm d) 127 mm
- 17) A length of glass tubing is 0.525m. How many inches long is that tubing?
 $2.54\text{cm} = 1\text{ in}$
 a) 0.0133 in b) 20.7 in c) 52.5 in d) 133 in
- 18) Rutherford carried out experiments in which a beam of alpha particles was directed at a thin piece of metal foil. From these experiments he concluded that:
 (a) Electrons are massive particles.
 (b) The positively charged parts of atoms are moving about with a velocity approaching the speed of light.
 (c) The positively charged parts of atoms are extremely small and extremely heavy particles.
 (d) The diameter of an electron is approximately equal to that of the nucleus.
 (e) Electrons travel in circular orbits around the nucleus.
- 19) If you do not understand a direction or part of a lab procedure, you should:
 a) figure it out as you do the lab
 b) ask the instructor for clarification before proceeding
 c) try several methods until something works
 d) skip it and go on to the next part
19. What is the name of experiment used by William Crookes to determine charge of electrons?
 a) Cathode rays scattering experiment b) Alpha particles scattering experiment
 c) Beta particles scattering experiment d) An oil drop experiment
20. One false about cathode rays:
 a) Their properties are independent of the electrodes, and the gas present in the cathode ray tube.
 b) The charge/mass ratio of the rays is constant.
 c) Upon passing through electric field, the cathode rays bend towards the negative plate
 d) Cathode rays are negatively charged and affected by magnetic field

21. The Law of Definite Proportions states that:
- Atoms are identical in mass.
 - Elements can combine in different ratios.
 - A chemical compound always contains its elements in a fixed ratio by mass.
 - Atoms are indivisible.
22. The modern view of atomic structure includes:
- Atoms being solid spheres.
 - Atoms being indivisible.
 - Electrons in fixed orbits around the nucleus.
 - Electrons existing in probabilistic electron clouds

III. Match those chemists I column “A” and philosophers with their basic conceptual statements in column “B” items.

<u>A</u>	<u>B</u>
1) Democritus	A. Both matter and mass are neither created nor destroyed.
2) Empedocles	B. indivisible and indestructible atoms are the basic components of all matter in the universe
3) Antoine Lavoisier	C. There is a void, which is an empty space between atoms.
4) John Dalton	D. every chemical compound contains fixed and constant proportions (by mass) of its constituent elements
5) Epicurus	E. All materials are made up of four things called elements: earth, air, water, and fire.
	F. All atoms of the same element are identical and have the same mass and size.
	G. Substances are infinitely divisible.

IV. Fill in the blanks.

- The presence of positively charged particles in an atom has been predicted by_____.
- _____ is an explanation of an aspect of the natural world and universe that can be repeatedly tested and verified by the scientific method.
- _____states that every chemical compound contains fixed and constant proportions (by mass) of its constituent elements

V. Write in short

- Write at least 2 drawbacks and feedback based statements of Dalton Atomic Theory.-----2 points
- List out the three laws of chemical reaction with their full statement. -----3 points.

Chemistry Answer Sheet Grade 9

Full Name _____ Grade & Section 9 R.No _____
Subject:- Chemistry Year:- 2018 E.C Exam Room _____ Time allowed:- 2:00'

I. True / False

1. _____
2. _____
3. _____
4. _____
5. _____

II Choose

1. _____
2. _____
3. _____
4. _____
5. _____
6. _____
7. _____
8. _____
9. _____

10. _____
11. _____
12. _____
13. _____
14. _____
15. _____
16. _____
17. _____
18. _____
19. _____

20. _____
21. _____
22. _____

III Matching

1. _____
2. _____
3. _____
4. _____
5. _____

IV. Fill in the blanks.

1. _____
2. _____
3. _____

V: - Write in short

1.
A. _____
B. _____
2.
A. _____
B. _____
C. _____